

Lab Continuation: Wazuh SIEM Deployment and Agent Integration

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Wazuh 4.7.5 · Ubuntu VM · Windows Server 2022 · VirtualBox · CIS Benchmark

Context: This section continues from the previous lab report, which documented the reconnaissance, brute-force simulation, and native log-based detection phases, ending with two failed Wazuh installation attempts due to hardware constraints. In a second session, the deployment was completed successfully and the Windows Server 2022 agent was connected and confirmed active.

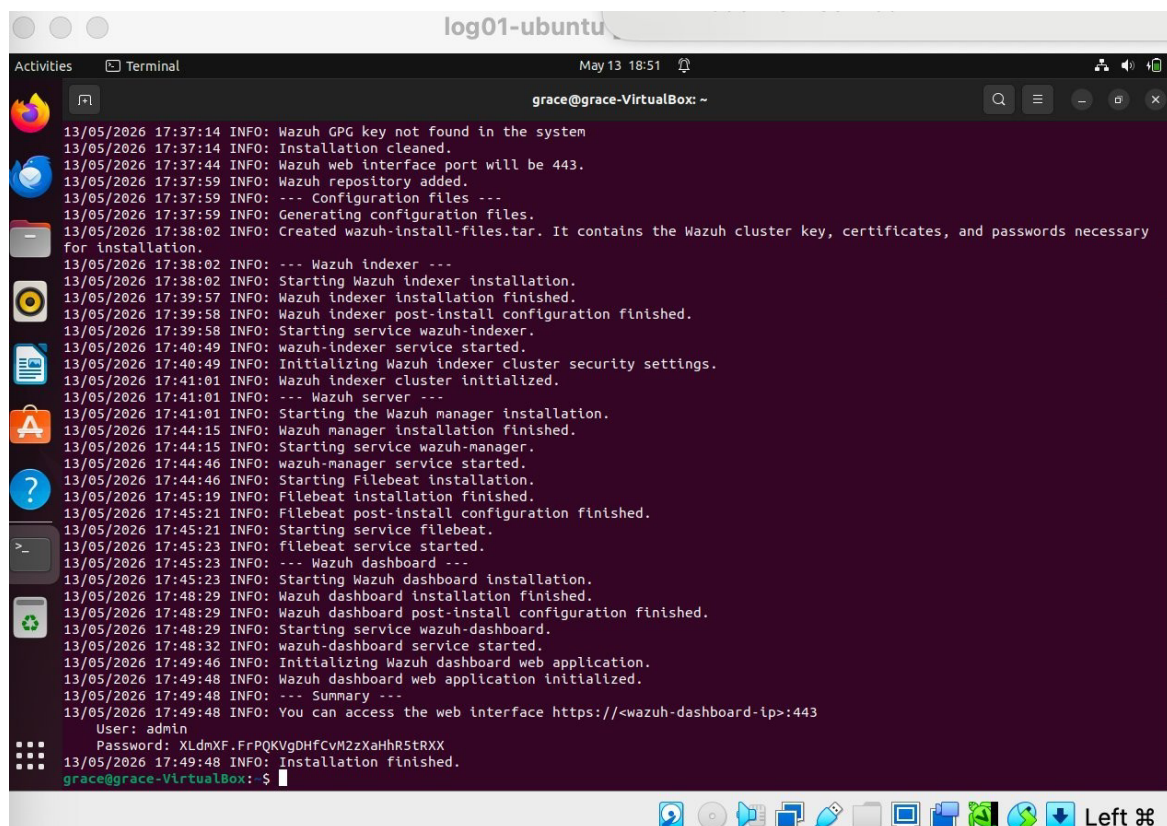
Wazuh SIEM Deployment — Third Attempt

Resolving the Hardware Constraint

The previous two Wazuh installation attempts failed at different stages. The first hit the 4 GB RAM minimum requirement wall immediately. The second, run with the `-i` bypass flag, got through the full installation sequence before the `wazuh-indexer` service failed to start and the installer self-cleaned. After both failures, the root cause was clear: the Ubuntu VM did not have enough RAM to start the indexer reliably, and the `-i` flag does not actually fix that.

The fix required changing the environment rather than the command. The Windows Server VM was shut down completely to free available memory on the host, and the Ubuntu VM allocation was increased to 4 GB in VirtualBox settings before starting the install again. When the installer ran this time, it flagged that leftover files from the previous failed attempt still existed and refused to proceed. Running the script with the `--overwrite` flag cleared those files and the installation ran through without interruption.

What this showed: Reading the installer output carefully each time made the next step obvious. The hardware error, the indexer failure, the leftover file conflict: each one had a specific message that pointed directly to the cause. None of them required guessing.



```
log01-ubuntu
Activities Terminal May 13 18:51
grace@grace-VirtualBox: ~
13/05/2026 17:37:14 INFO: Wazuh GPG key not found in the system
13/05/2026 17:37:14 INFO: Installation cleaned.
13/05/2026 17:37:44 INFO: Wazuh web interface port will be 443.
13/05/2026 17:37:59 INFO: Wazuh repository added.
13/05/2026 17:37:59 INFO: --- Configuration files ---
13/05/2026 17:37:59 INFO: Generating configuration files.
13/05/2026 17:38:02 INFO: Created wazuh-install-files.tar. It contains the Wazuh cluster key, certificates, and passwords necessary
for installation.
13/05/2026 17:38:02 INFO: --- Wazuh indexer ---
13/05/2026 17:38:02 INFO: Starting Wazuh indexer installation.
13/05/2026 17:39:57 INFO: Wazuh indexer installation finished.
13/05/2026 17:39:58 INFO: Wazuh indexer post-install configuration finished.
13/05/2026 17:39:58 INFO: Starting service wazuh-indexer.
13/05/2026 17:40:49 INFO: wazuh-indexer service started.
13/05/2026 17:40:49 INFO: Initializing Wazuh indexer cluster security settings.
13/05/2026 17:41:01 INFO: Wazuh indexer cluster initialized.
13/05/2026 17:41:01 INFO: --- Wazuh server ---
13/05/2026 17:41:01 INFO: Starting the Wazuh manager installation.
13/05/2026 17:44:15 INFO: Wazuh manager installation finished.
13/05/2026 17:44:15 INFO: Starting service wazuh-manager.
13/05/2026 17:44:46 INFO: wazuh-manager service started.
13/05/2026 17:44:46 INFO: Starting Filebeat installation.
13/05/2026 17:45:19 INFO: Filebeat installation finished.
13/05/2026 17:45:21 INFO: Filebeat post-install configuration finished.
13/05/2026 17:45:21 INFO: Starting service filebeat.
13/05/2026 17:45:23 INFO: filebeat service started.
13/05/2026 17:45:23 INFO: --- Wazuh dashboard ---
13/05/2026 17:45:23 INFO: Starting Wazuh dashboard installation.
13/05/2026 17:48:29 INFO: Wazuh dashboard installation finished.
13/05/2026 17:48:29 INFO: Wazuh dashboard post-install configuration finished.
13/05/2026 17:48:29 INFO: Starting service wazuh-dashboard.
13/05/2026 17:48:32 INFO: wazuh-dashboard service started.
13/05/2026 17:49:46 INFO: Initializing Wazuh dashboard web application.
13/05/2026 17:49:48 INFO: Wazuh dashboard web application initialized.
13/05/2026 17:49:48 INFO: --- Summary ---
13/05/2026 17:49:48 INFO: You can access the web interface https://<wazuh-dashboard-ip>:443
User: admin
Password: XldmXF.FrPQKVgDHfCvW2zXaHhR5tRXX
13/05/2026 17:49:48 INFO: Installation finished.
grace@grace-VirtualBox: $
```

Figure 6. Wazuh re-installation with `--overwrite` flag resolving leftover file conflict, completing all services successfully.

Confirming the Installation

The completed installation summary confirmed the dashboard was available at <https://192.168.56.2> on port 443, with admin credentials generated during setup. Opening the Wazuh dashboard in Firefox on the Ubuntu VM and navigating to <https://192.168.56.2> (without the /24 subnet notation from the ip a output, which caused an initial 401 error) brought up the health check page. Four of five checks passed immediately. The API version check returned a timeout warning, which resolved on its own as the manager service finished initialising.

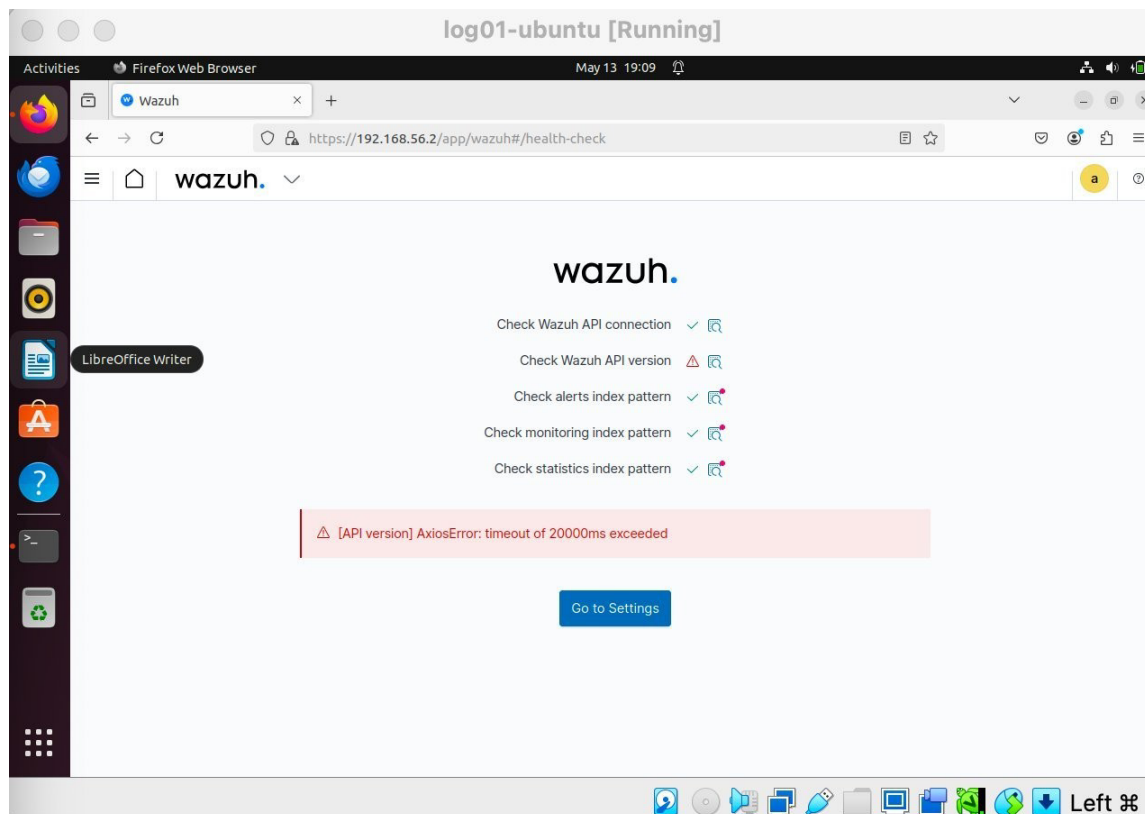


Figure 7. Wazuh dashboard health check showing API connection, alerts index, monitoring index, and statistics index all passing.

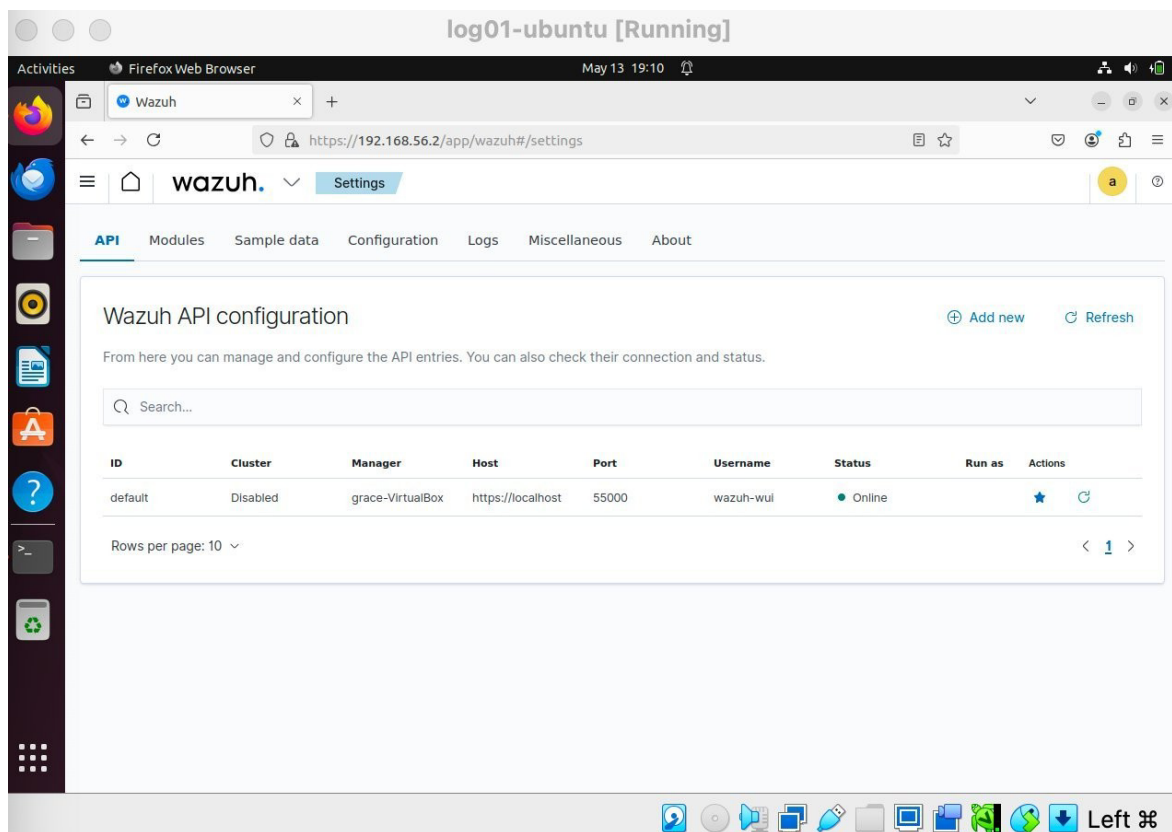


Figure 8. Wazuh API configuration confirming the grace-VirtualBox manager online on port 55000.

Windows Server 2022 Agent Deployment

Getting the Install Command onto the Target

Clipboard sharing between the Mac host and VirtualBox VMs was not functioning throughout the lab, which made transferring commands between the Ubuntu and Windows Server VMs a recurring friction point. For the agent deployment, rather than trying to copy the generated PowerShell command across VMs, the Wazuh dashboard was accessed directly from inside the Windows Server VM using Edge. Navigating to <https://192.168.56.2>, accepting the certificate warning, and logging in with the admin credentials brought up the same dashboard. From there, the Deploy new agent wizard was used to select Windows MSI and enter the server address, and the generated command was copied and pasted within the Windows Server VM itself, which bypassed the cross-VM clipboard issue entirely.

Why this mattered: Small environmental constraints compound in a home lab. Finding a clean path around the clipboard issue rather than spending time fighting it kept the session moving.

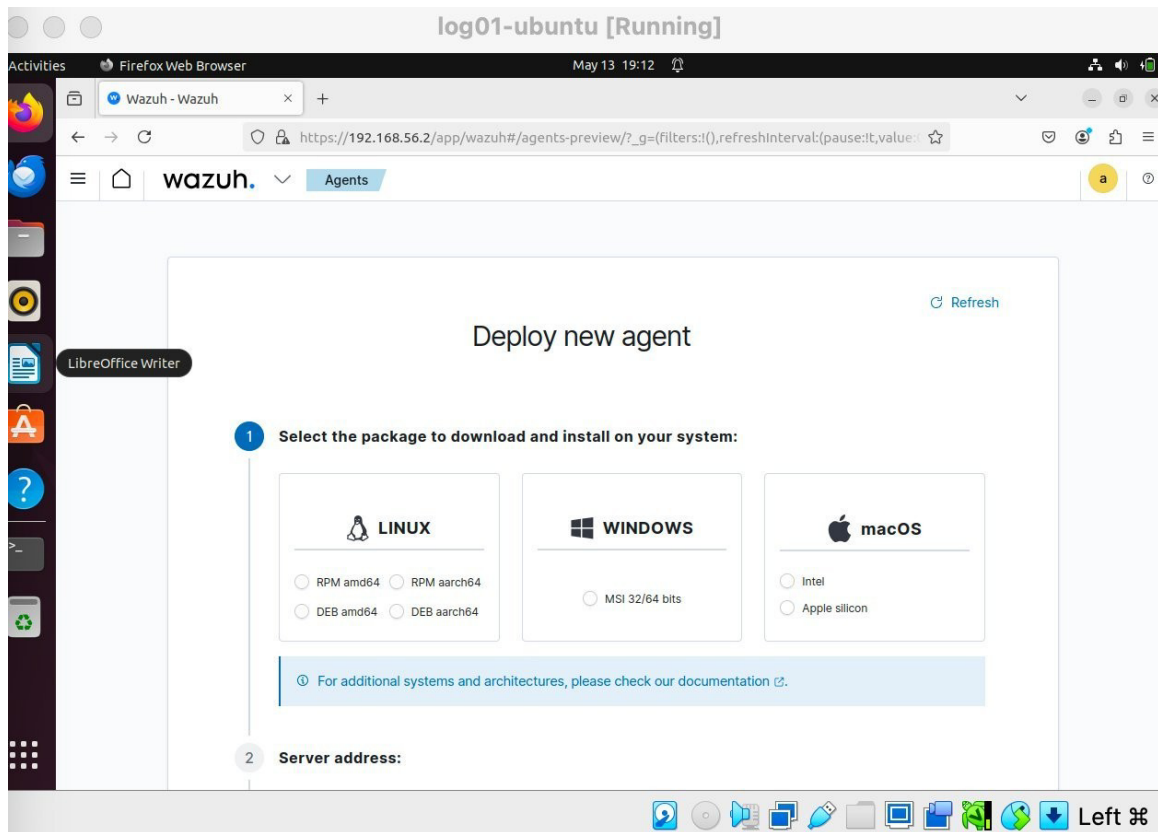
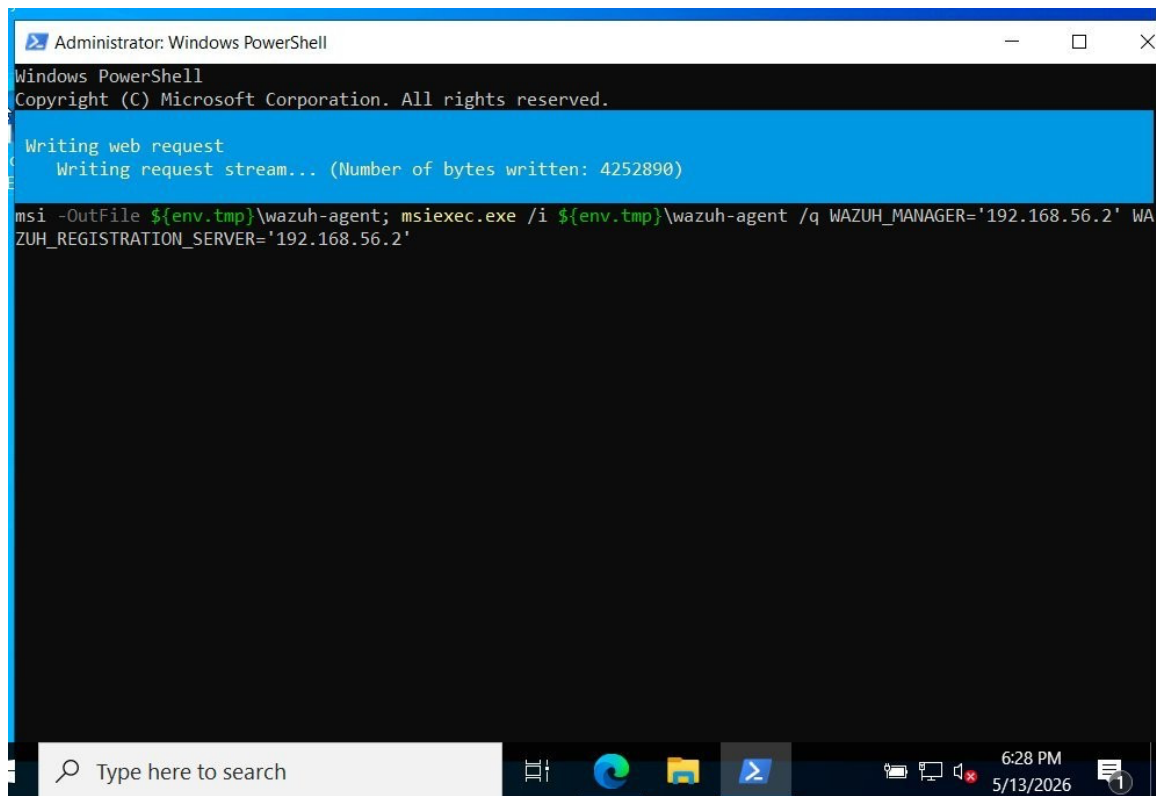


Figure 9. Wazuh Deploy new agent wizard inside the Windows Server VM browser, showing OS selection and server address configuration.

Installing the Agent

The generated PowerShell command downloaded the Wazuh agent MSI from packages.wazuh.com, approximately 40 MB, and ran the installer silently with the Wazuh manager address pre-configured. The download took about three minutes. The installation completed without errors and returned to the PowerShell prompt cleanly. The WazuhSvc service was then started with `NET START WazuhSvc`.

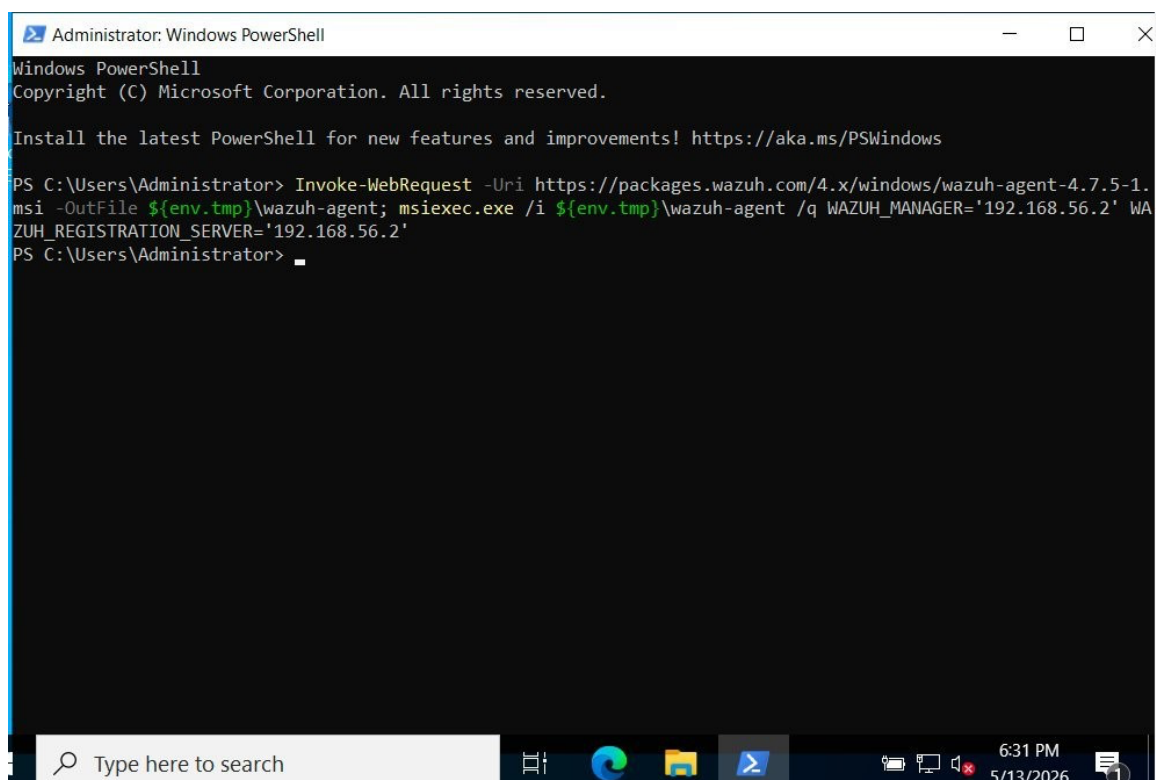


```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Writing web request
Writing request stream... (Number of bytes written: 4252890)

msi -OutFile ${env.tmp}\wazuh-agent; msixec.exe /i ${env.tmp}\wazuh-agent /q WAZUH_MANAGER='192.168.56.2' WAZUH_REGISTRATION_SERVER='192.168.56.2'
```

Figure 10. PowerShell downloading the Wazuh agent MSI on Windows Server 2022, showing bytes written during transfer.



```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Administrator> Invoke-WebRequest -Uri https://packages.wazuh.com/4.x/windows/wazuh-agent-4.7.5-1.msi -OutFile ${env.tmp}\wazuh-agent; msixec.exe /i ${env.tmp}\wazuh-agent /q WAZUH_MANAGER='192.168.56.2' WAZUH_REGISTRATION_SERVER='192.168.56.2'
PS C:\Users\Administrator>
```

Figure 11. Agent install command completing cleanly with no errors and returning to the prompt.

Agent Active and Automated CIS Benchmark Scan

Switching back to the Wazuh dashboard on the Ubuntu VM and refreshing the Agents page showed WIN-EEJ1IO99076 registered at 192.168.56.4, running Windows Server 2022 Datacenter Evaluation, Wazuh agent version v4.7.5, status active. Agent coverage showed 100 percent. The full deployment, from starting the install command to seeing the agent confirmed active, took under five minutes.

What was unexpected was what happened next without any additional configuration. Wazuh automatically initiated a Security Configuration Assessment against the CIS Microsoft Windows Server 2022 Benchmark v1.0.0 the moment the agent connected. The scan completed at 19:38 and returned a compliance score of 35 percent: 120 controls passing, 216 failing, and 6 not applicable across 342 evaluated controls. That is a full automated posture assessment of the Windows Server environment, produced immediately and without any manual input.

Security relevance: A 35 percent CIS compliance score on a default Windows Server 2022 installation reflects the real gap between out-of-the-box configuration and hardened production standards. Having that data surface automatically the moment an agent registers is a capability that changes what SIEM monitoring means in practice. It is not just event aggregation: it is continuous compliance visibility from the first connection.

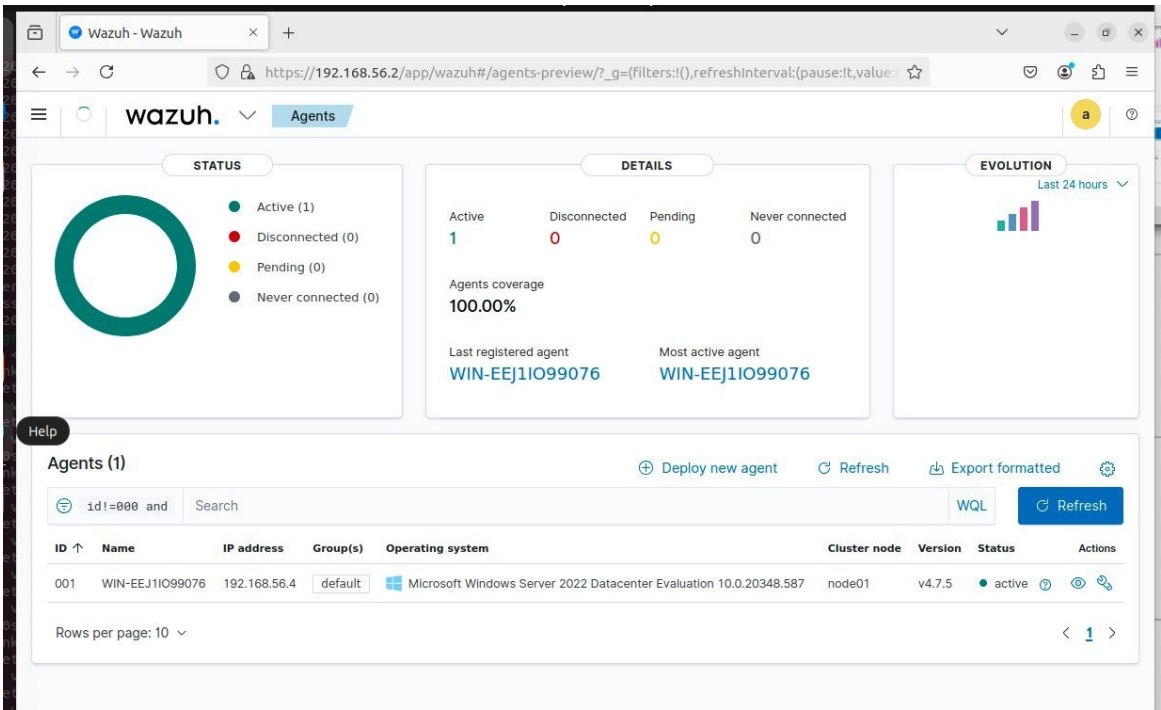


Figure 12. Wazuh Agents dashboard showing WIN-EEJ1IO99076 active at 192.168.56.4 with 100 percent agent coverage.

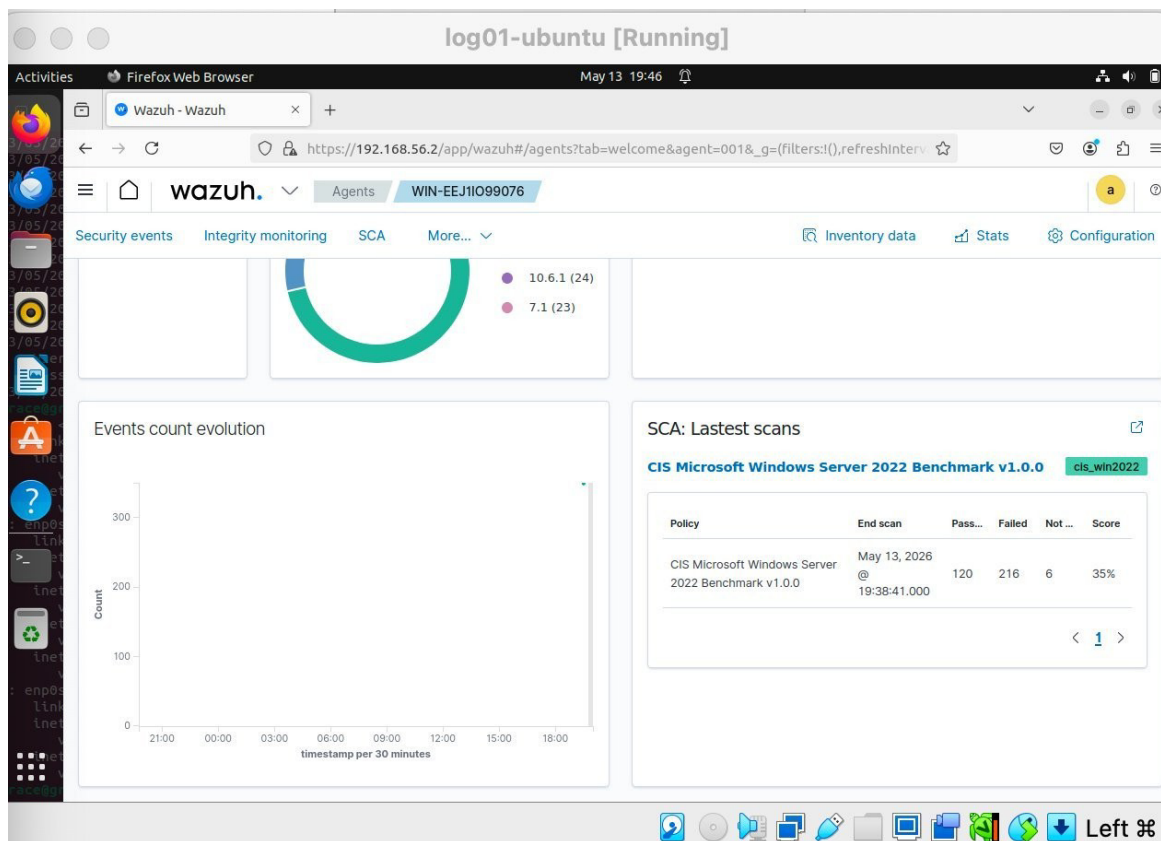


Figure 13. Wazuh agent detail for WIN-EEJ1IO99076 showing events count evolution and CIS Benchmark scan result: 35 percent compliance, 120 passed, 216 failed.